

MATH-1014 T10B

Tutorial 9: Improper Integrals (with Advanced Tests)

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Today's Roadmap

1 Concept Review: Improper Integrals

2 Practice Problems & Solutions

Review: Proper vs. Improper Integrals

Before diving into calculations, we must distinguish between **Proper** and **Improper** integrals for $\int_a^b f(x)dx$.

Proper Integrals (The Good Ones)

- The interval $[a, b]$ is a **finite interval**.
- The function $f(x)$ is **continuous and bounded** on $[a, b]$.
- Result: Always evaluates to a finite number (Fundamental Theorem of Calculus applies directly).

Improper Integrals (Require Limits)

- **Type I:** The interval is infinite (e.g., $[a, \infty)$, $(-\infty, b]$, or $(-\infty, \infty)$).
- **Type II:** The integrand $f(x)$ has an infinite discontinuity (vertical asymptote) somewhere in $[a, b]$.

Review: Type I Improper Integrals

For infinite intervals, we replace the infinity symbol with a variable t , evaluate the definite integral, and then take the limit as $t \rightarrow \infty$.

Mathematical Definitions

- ① $\int_a^\infty f(x)dx = \lim_{t \rightarrow \infty} \int_a^t f(x)dx$
- ② $\int_{-\infty}^b f(x)dx = \lim_{t \rightarrow -\infty} \int_t^b f(x)dx$
- ③ $\int_{-\infty}^\infty f(x)dx = \int_{-\infty}^c f(x)dx + \int_c^\infty f(x)dx$ (Choose any c , usually 0)

Terminology: If the limit exists as a finite number, the integral is **convergent**. If the limit does not exist (or equals $\pm\infty$), the integral is **divergent**. (Note: For case 3, **both** parts must converge for the whole integral to converge).

Important Shortcut: The p-Test for Type I

A very common family of integrals involves $\frac{1}{x^p}$. You should memorize their convergence behavior.

The p-Test Theorem

The improper integral

$$\int_1^{\infty} \frac{1}{x^p} dx$$

- **Converges** if $p > 1$ (The curve approaches the x-axis fast enough).
- **Diverges** if $p \leq 1$ (The curve approaches the x-axis too slowly, enclosing infinite area).

Example: $\int_1^{\infty} \frac{1}{x^2} dx = 1$ (Convergent, $p = 2$), but $\int_1^{\infty} \frac{1}{x} dx = \infty$ (Divergent, $p = 1$).

Review: Type II Improper Integrals

If $f(x)$ is discontinuous at a point, we must approach that point using a one-sided limit.

Handling Discontinuities

- **Discontinuous at upper limit b :**

$$\int_a^b f(x)dx = \lim_{t \rightarrow b^-} \int_a^t f(x)dx$$

- **Discontinuous at lower limit a :**

$$\int_a^b f(x)dx = \lim_{t \rightarrow a^+} \int_t^b f(x)dx$$

The Hidden Trap: Interior Discontinuity

If $f(x)$ has a discontinuity at c where $a < c < b$, you **cannot** integrate directly from a to b . You **must** split it:

$$\int_a^b f(x)dx = \int_a^c f(x)dx + \int_c^b f(x)dx$$

Review: The Direct Comparison Theorem

Sometimes, finding the exact antiderivative is impossible (e.g., e^{-x^2}). Instead, we compare it to a simpler function. **Note: Functions must be positive!**

Comparison Test (for $f(x) \geq g(x) \geq 0$)

- **Big converges $\implies$ Small converges:**
If $\int_a^\infty f(x)dx$ converges, then $\int_a^\infty g(x)dx$ converges.
- **Small diverges $\implies$ Big diverges:**
If $\int_a^\infty g(x)dx$ diverges, then $\int_a^\infty f(x)dx$ diverges.

Challenge: What if $f(x)$ looks like $\frac{1}{x^2-0.5}$? It's **larger** than $\frac{1}{x^2}$ (which converges), so the Direct Comparison Test fails to give a conclusion!

Deeper Tool: The Limit Comparison Test (LCT)

When direct inequalities are hard to build, we analyze the **dominant terms** as $x \rightarrow \infty$.

Limit Comparison Test

Suppose $f(x) > 0$ and $g(x) > 0$. If

$$\lim_{x \rightarrow \infty} \frac{f(x)}{g(x)} = L$$

where $0 < L < \infty$ (a positive, finite number), then $\int_a^\infty f(x)dx$ and $\int_a^\infty g(x)dx$ **share the same fate** (either both converge or both diverge).

Strategy: For a rational function, drop the lower-degree terms to find $g(x)$.

E.g., $f(x) = \frac{3x^2+5}{x^5-2x+1}$. As $x \rightarrow \infty$, $f(x) \approx \frac{3x^2}{x^5} = \frac{3}{x^3}$. Choose $g(x) = \frac{1}{x^3}$.

Problem 1: Type I with Partial Fractions

Problem 1.

Evaluate the following improper integral or show that it diverges:

$$\int_0^{\infty} \frac{1}{x^2 + 3x + 2} dx$$

Hint: A polynomial in the denominator should remind you of Partial Fraction Decomposition. Be careful with how you evaluate the limits of logarithmic terms at infinity.

Solution 1

Step 1: Set up the limit and factor the denominator.

$$\int_0^{\infty} \frac{1}{(x+1)(x+2)} dx = \lim_{t \rightarrow \infty} \int_0^t \frac{1}{(x+1)(x+2)} dx$$

Step 2: Partial Fraction Decomposition.

$$\frac{1}{(x+1)(x+2)} = \frac{A}{x+1} + \frac{B}{x+2}$$

Solving this yields $A = 1$ and $B = -1$. So the integral becomes:

$$\lim_{t \rightarrow \infty} \int_0^t \left(\frac{1}{x+1} - \frac{1}{x+2} \right) dx$$

Step 3: Integrate.

$$= \lim_{t \rightarrow \infty} [\ln |x+1| - \ln |x+2|]_0^t$$

Solution 1 (Continued)

Step 4: Combine logs before taking the limit.

If we plug in t directly, we get $\infty - \infty$ (an indeterminate form). We must use logarithm rules to combine them:

$$\ln|x+1| - \ln|x+2| = \ln\left|\frac{x+1}{x+2}\right|$$

Now evaluate from 0 to t :

$$\begin{aligned} I &= \lim_{t \rightarrow \infty} \left(\ln\left|\frac{t+1}{t+2}\right| - \ln\left|\frac{0+1}{0+2}\right| \right) \\ &= \lim_{t \rightarrow \infty} \ln\left(\frac{1+1/t}{1+2/t}\right) - \ln\left(\frac{1}{2}\right) \\ &= \ln(1) - \ln(0.5) \\ &= 0 - (-\ln 2) = \ln 2 \end{aligned}$$

The integral **converges** to $\ln 2$.

Problem 2: Type II with Integration by Parts

Problem 2.

Evaluate the following improper integral or show that it diverges:

$$\int_0^1 \frac{\ln(x)}{\sqrt{x}} dx$$

Hint: This is an endpoint singularity. You will need Integration by Parts to find the antiderivative, and L'Hôpital's Rule to resolve a $0 \cdot (-\infty)$ limit.

Solution 2

Step 1: Identify the singularity and set up the limit.

The integrand has a vertical asymptote as $x \rightarrow 0^+$.

$$\int_0^1 x^{-1/2} \ln(x) dx = \lim_{t \rightarrow 0^+} \int_t^1 x^{-1/2} \ln(x) dx$$

Step 2: Integration by Parts.

Let $u = \ln(x) \implies du = \frac{1}{x} dx$.

Let $dv = x^{-1/2} dx \implies v = 2x^{1/2} = 2\sqrt{x}$.

$$\begin{aligned} \int x^{-1/2} \ln(x) dx &= 2\sqrt{x} \ln(x) - \int 2\sqrt{x} \left(\frac{1}{x} \right) dx \\ &= 2\sqrt{x} \ln(x) - 2 \int x^{-1/2} dx \\ &= 2\sqrt{x} \ln(x) - 4\sqrt{x} \end{aligned}$$

Solution 2 (Continued)

Step 3: Evaluate limits.

$$\begin{aligned}\lim_{t \rightarrow 0^+} [2\sqrt{x} \ln(x) - 4\sqrt{x}]_t^1 &= (2\sqrt{1} \ln(1) - 4\sqrt{1}) - \lim_{t \rightarrow 0^+} (2\sqrt{t} \ln(t) - 4\sqrt{t}) \\ &= (0 - 4) - 2 \lim_{t \rightarrow 0^+} (\sqrt{t} \ln(t)) + 0\end{aligned}$$

Step 4: L'Hôpital's Rule for $\lim_{t \rightarrow 0^+} \sqrt{t} \ln(t)$.

This is $0 \cdot (-\infty)$. Rewrite as $\frac{-\infty}{\infty}$:

$$\lim_{t \rightarrow 0^+} \frac{\ln(t)}{t^{-1/2}} \stackrel{\text{L'H}}{=} \lim_{t \rightarrow 0^+} \frac{1/t}{-\frac{1}{2}t^{-3/2}} = \lim_{t \rightarrow 0^+} (-2t^{1/2}) = 0$$

Conclusion:

$$I = -4 - 2(0) = -4$$

The integral **converges** to -4 .

Problem 3: The Interior Singularity (Convergent Case)

Problem 3.

Evaluate the following improper integral if possible:

$$\int_{-1}^8 \frac{1}{\sqrt[3]{x}} dx$$

Warning

Do not blindly apply the Fundamental Theorem of Calculus! Where is the integrand undefined?

Solution 3

Step 1: Identify the hidden discontinuity.

The integrand $f(x) = x^{-1/3}$ is discontinuous at $x = 0$, which is **inside** the interval $[-1, 8]$. We **MUST** split the integral into two separate one-sided limits:

$$\int_{-1}^8 x^{-1/3} dx = \lim_{t \rightarrow 0^-} \int_{-1}^t x^{-1/3} dx + \lim_{s \rightarrow 0^+} \int_s^8 x^{-1/3} dx$$

Step 2: Find the antiderivative.

$$\int x^{-1/3} dx = \frac{3}{2} x^{2/3}$$

Step 3: Evaluate the left segment.

$$\lim_{t \rightarrow 0^-} \left[\frac{3}{2} x^{2/3} \right]_{-1}^t = \lim_{t \rightarrow 0^-} \left(\frac{3}{2} t^{2/3} - \frac{3}{2} (-1)^{2/3} \right) = 0 - \frac{3}{2} (1) = -\frac{3}{2}$$

Solution 3 (Continued)

Step 4: Evaluate the right segment.

$$\lim_{s \rightarrow 0^+} \left[\frac{3}{2} x^{2/3} \right]_s^8 = \lim_{s \rightarrow 0^+} \left(\frac{3}{2} (8)^{2/3} - \frac{3}{2} s^{2/3} \right)$$

Since $8^{2/3} = (\sqrt[3]{8})^2 = 2^2 = 4$:

$$= \frac{3}{2}(4) - 0 = 6$$

Conclusion:

Since **both** one-sided limits converge to finite numbers, the whole integral converges! We just add them together:

$$\int_{-1}^8 \frac{1}{\sqrt[3]{x}} dx = -\frac{3}{2} + 6 = \frac{9}{2} \text{ (or 4.5)}$$

Note: Not all interior singularities lead to divergence! It depends on how "fast" the function goes to infinity (p-test rules still intuitively apply here).

Problem 4: When LCT Fails (Direct Comparison Test)

Problem 4.

Use the Direct Comparison Theorem to determine whether the following integral converges or diverges:

$$\int_1^{\infty} \frac{3 + \cos(x)}{x^2 + \sqrt{x}} dx$$

Conceptual Question

Why can't we use the Limit Comparison Test (LCT) by comparing it to $g(x) = \frac{1}{x^2}$?

Hint: Think about what happens to $\cos(x)$ as $x \rightarrow \infty$.

Solution 4

Step 1: Why LCT fails here.

If we try LCT with $g(x) = \frac{1}{x^2}$, we need to evaluate $\lim_{x \rightarrow \infty} \frac{3 + \cos(x)}{1 + x^{-1.5}}$. Because $\cos(x)$ oscillates constantly between -1 and 1 , **this limit does not exist**. Therefore, LCT is useless here. We **must** use the Direct Comparison Test (DCT).

Step 2: Construct the inequality (Bounding).

To show a fraction is smaller than something else (to prove convergence), we must **maximize the numerator** and **minimize the denominator**.

- **Numerator:** Since $-1 \leq \cos(x) \leq 1$, the maximum value of $3 + \cos(x)$ is 4.
- **Denominator:** For $x \geq 1$, adding $\sqrt{x}$ makes the denominator larger. So, $x^2 + \sqrt{x} > x^2$.

Combining these two facts, for all $x \geq 1$:

$$0 \leq \frac{3 + \cos(x)}{x^2 + \sqrt{x}} \leq \frac{4}{x^2}$$

Solution 4 (Continued)

Step 3: Evaluate the comparison function.

We now look at the integral of our upper bound:

$$\int_1^{\infty} \frac{4}{x^2} dx = 4 \int_1^{\infty} \frac{1}{x^2} dx$$

This is a standard p -integral with $p = 2$. Since $p > 1$, this integral **converges**.

Step 4: Conclusion.

Because our original function is strictly smaller than a function whose integral converges (and both are positive), our integral must also converge by the **Direct Comparison Theorem**.

$$\int_1^{\infty} \frac{3 + \cos(x)}{x^2 + \sqrt{x}} dx \quad \text{Converges.}$$

Problem 5: Proving Divergence with DCT

Problem 5.

Use the Direct Comparison Theorem to determine whether the following integral converges or diverges:

$$\int_2^{\infty} \frac{x^2 + 1}{\sqrt{x^6 - 1}} dx$$

Strategy Tip

- 1. Guess the behavior:** As $x \rightarrow \infty$, what are the dominant terms? Does it look like it converges or diverges?
- 2. Build the bound:** If you suspect it diverges, you must find a simpler function that is strictly **smaller**. How do you make a fraction smaller?

Solution 5

Step 1: Predict the outcome.

For very large x , the $+1$ and -1 matter very little. The function behaves roughly like $\frac{x^2}{\sqrt{x^6}} = \frac{x^2}{x^3} = \frac{1}{x}$. Since $\int \frac{1}{x} dx$ diverges, we suspect our integral **diverges**.

Step 2: Construct the inequality (Bounding DOWN).

To prove divergence using DCT, we must show our function is **larger** than a divergent function. We need to make the original fraction **smaller**:

- **Minimize the Numerator:** Dropping $+1$ makes it smaller. So, $x^2 + 1 > x^2$.
- **Maximize the Denominator:** Dropping -1 makes the inside of the root larger, which makes the whole denominator larger. So, $\sqrt{x^6 - 1} < \sqrt{x^6} = x^3$.

Solution 5 (Continued)

Step 3: Combine the bounds.

Since the numerator is strictly larger than x^2 , and the denominator is strictly smaller than x^3 , the overall fraction must be larger than their ratio:

$$\frac{x^2 + 1}{\sqrt{x^6 - 1}} > \frac{x^2}{x^3} = \frac{1}{x} > 0$$

(Note: This holds true for all $x \geq 2$)

Step 4: Evaluate the comparison function and conclude.

We know that the integral of our lower bound:

$$\int_2^{\infty} \frac{1}{x} dx$$

diverges (p-test with $p = 1 \leq 1$).

Since our original function is strictly **larger** than a divergent function, it must also diverge by the **Direct Comparison Theorem**.

$$\int_2^{\infty} \frac{x^2 + 1}{\sqrt{x^6 - 1}} dx \quad \text{Diverges.}$$

THANK YOU!

